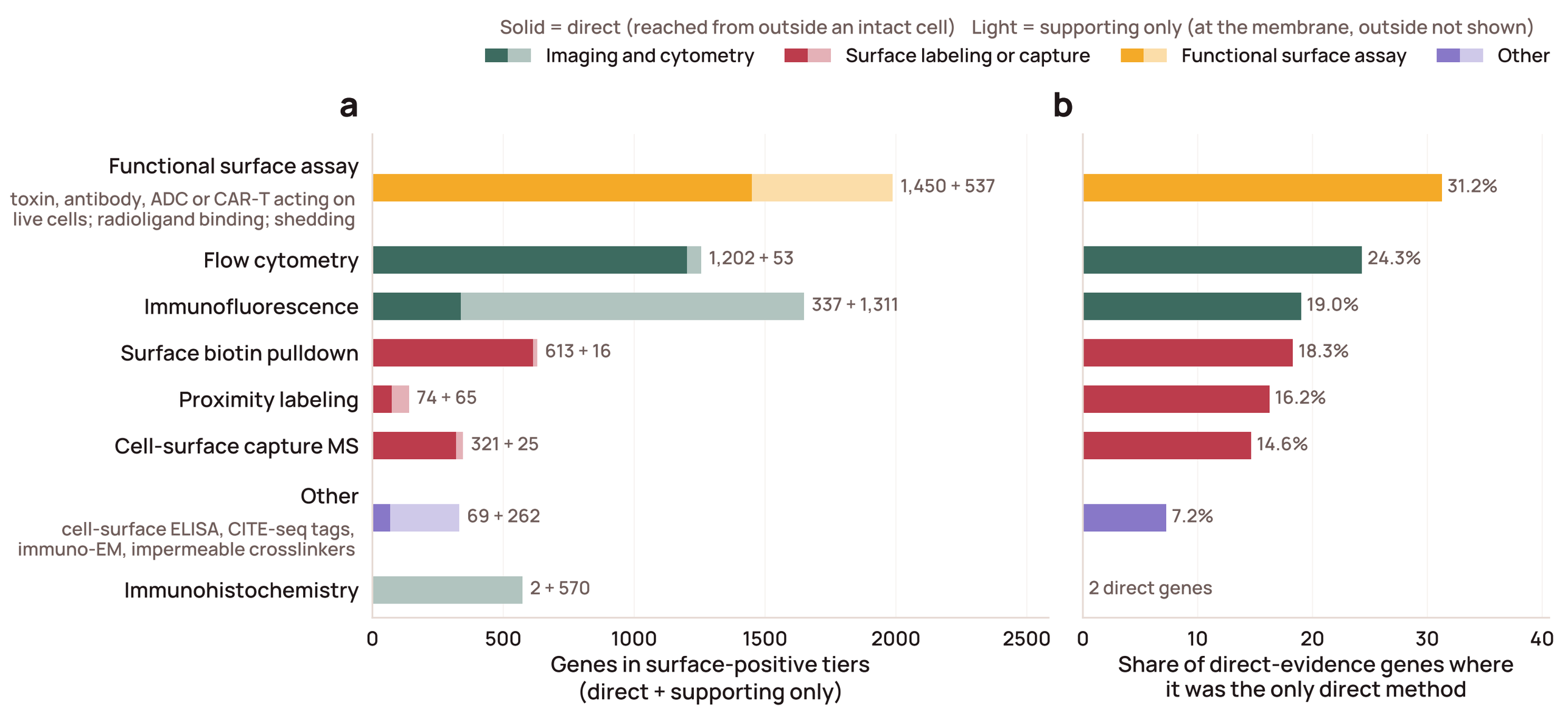
Solid = direct (reached from outside an intact cell)
Light = supporting only (at the membrane, outside not shown)
Imaging and cytometry
Surface labeling or capture
Functional surface assay
Other
a
b
Functional surface assay
toxin, antibody, ADC or CAR-T acting on live cells; radioligand binding; shedding
Flow cytometry
Immunofluorescence
Surface biotin pulldown
Proximity labeling
Cell-surface capture MS
Other
cell-surface ELISA, CITE-seq tags, immuno-EM, impermeable crosslinkers
Immunohistochemistry
1,450 + 537
1,202 + 53
337 + 1,311
613 + 16
74 + 65
321 + 25
69 + 262
2 + 570
0
500
1000
1500
2000
2500
Genes in surface-positive tiers (direct + supporting only)
31.2%
24.3%
19.0%
18.3%
16.2%
14.6%
7.2%
2 direct genes
0
10
20
30
40
Share of direct-evidence genes where it was the only direct method